Introduction to Machine Learning

HW2: Classify a Music Catelog

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HMI

Rules

- Individual assignment. No plagiarism.
- All kinds of AI tools are permitted. But you must submit the AI-use log.
- **Do not use AI tools to generate the content of the report.**
- Report can be either written in English or Traditional Chinese.
- No GPU is required. Google Colab is sufficient enough to complete the task.
- Do not use external datasets, features, or information collected from other sources.
- Submit your work on eeclass. **No late submission will be accepted.**

Deadline

Submission	Deadline
A **pre-registration report** in the PDF format, up to **2 pages.** File name: HW2_pre_report_{student_id}.pdf	10/20 11:59 PM
A **final report** in the PDF format, up to **15 pages.** File name: HW2_final_report_{student_id}.pdf	11/3 11:59 PM
An **.ipynb notebook** containing all code needed to reproduce the results reported in your submission. File name: HW2_code_{student_id}.ipynb	
The **AI-use log** document. File name: HW2_ai_use_{student_id}.md	

About AI Use Log

- Submit an AI-use log written in the markdown file:
  - What you used AI for.
  - How you checked the output of AI.
- You should be able to explain any AI-assisted work during the demo session.
- A 5-point deduction will be applied for missing submissions or conspicuously low-effort responses (e.g., merely writing "I used ChatGPT").

Materials

- We provide:
  - the starter notebook
  - The *processed* dataset
  - The build script for reference (reproduce the *processed* dataset from the raw one)
  - **Assigned reading: Model Selection and Evaluation**
- Please check out eeclass or course website for more details.
- Please check out the spec (which is in pdf format) for complete information about the homework.

Introduction

The team now wants to use the music catalog for a concrete purpose: **automatically tagging the genre of tracks that are added to the catalog in the future.** You are given a *processed* subset of the same catalog with **5,300 tracks from 20 genres** and asked to build, evaluate, and choose classifiers for the task.

- How does each classifier make decisions, and how do hyperparameters change what it learns?
- Why do different algorithms perform differently on the same data, and in which situations is each of them a good or bad choice?
- Is a single metric such as accuracy enough to judge a model performance, and if not, which metrics best align with the specific use cases?
- What does a reported score actually promise about tracks the model has never seen, and how should the test be designed so that the number is honest?

Purpose of the Homework

- Not to obtain the highest possible accuracy. You will not compete with your classmate!
- We expect you to:
  - **Explain the reasons** behind your analysis choices.
  - **Recognize** when the data support / not support the conclusion.
  - **Evaluate** each model **honestly** rather than evaluate through the flawed testing.
  - A weak or unsuccessful result may be a correct finding if you **demonstrate** it thoroughly.

Section 1: The Data



- Dataset of spotify tracks over a range of genres.
- We have already processed the dataset for you, so there is no need for auditing and preprocessing the dataset.
- Please check out the spec and script for more processing details.
- Contain 5,300 rows and 20 columns.

techno	ambient	comedy	blues	r-n-b
709	481	477	449	422
sleep	punk	jazz	hip-hop	classical
411	344	289	289	231
heavy-metal	bluegrass	reggae	country	grunge
223	205	160	141	118
death-metal	tango	opera	black-metal	salsa
100	82	62	58	49

- Definition of **rare genres**: the five smallest genres in the table **(death-metal, tango, opera, black-metal, and salsa)**
- **Imbalance data distribution**

Section 2: Pre-Registration (5%)

- **Submit this report before 10/20 11:59 PM. At most two pages.**
- **There are no formal answers!**
- **Build the evaluation protocol before you train any classifiers.**
- Read the assigned reading first, make sure you have fully understood the concept.
- Inside your report, you need to show:
  - Dataset Partitioning and Reproducibility
  - Leakage-Free Hyperparameter Tuning and Reporting Metrics
  - Empirical Justification of Stratification
  - Trivial Baselines
  - Theoretical Equivalence of Micro-F_1 and Accuracy

Some important evaluation techniques

- *k*-fold cross-validation
- Stratified sampling
- Nested (internal) cross-validation
- Leave-one-out cross-validation (LOOCV)
- Metrics for many classes: precision, recall, f1-score, and the macro and minor one.
- ROC, AUC, Confusion matrix

Section 3: Getting to Know Each Classifier (35%)

- Each classifier family introduced in class relies on distinct **inductive biases**, with **hyperparameters** controlling its decision dynamics.
- In this section, you will systematically **evaluate four classifier families**: Nearest-neighbor classifiers, Bayesian classifiers, decision trees and their ensembles, and support vector machines.
- Hint: **Library defaults** (e.g., in `scikit-learn`) **may not align with theoretical best practices or optimal convergence.** Identify, document, and justify any non-trivial default parameters you adjust or rely upon.

K-Nearest Neighbors (8%)

- **Look inside the prediction**
  - Train *k*-NN on the **standardized features**, analyze the nearest neighbors on the chosen tracks (which genre, the distance, and the vote).
- **The number of neighbors**
  - Vary *k* from 1 to **at least 100,** plot the training and cross-validated of accuracy and macro-F_1 against *k*.
  - Explain the curve, and which *k* is best by accuracy and by macro-f1. Explain why from how the votes work.
- **What does “near” mean**
  - Append *N* extra columns of noise drawn independently **from a standard normal distribution $\mathcal{N}(0, 1)$**. Choose several values of *N*.
  - **Explain the noise effect from the distance** that k-NN computes and examine the change of the nearest neighbors.

Naive Bayes (9%)

$$\log \frac{P(A \mid x)}{P(B \mid x)} = \log \frac{\pi_A}{\pi_B} + \sum_{j=1}^{13} \log \frac{\mathcal{N}(x_j; \mu_{Aj}, \sigma_{Aj}^2)}{\mathcal{N}(x_j; \mu_{Bj}, \sigma_{Bj}^2)},$$

- **Look inside the prediction**
  - Fit the classifier on the standardized features.
  - Pick one track *x* from the validation part, and the two genres *A* and *B* with the highest predicted probability for it.
  - **Compute the 14 terms included the prior term** from the fitted model's parameters.
  - Identify which terms decide the prediction.
- **A model that should not care about scale**
  - In principle rescaling a feature should not change its predictions.
  - Compare the classifier on raw and on standardized features. If the results differ, find out why from your library's documentation or source code.
  - **Vary the smoothing parameter** (`var_smoothing` in `scikit-learn`) over several orders of magnitude. The value you choose will be used in the rest of the homework.

Naive Bayes (9%)

- **When the assumptions are wrong**
  - Show at least one feature that clearly violates the **gaussian assumption**
  - Show at least one pair of features that clearly violates the **independence assumption**.
- **Confidence and the reliability diagram:**
  - Add exact copies of one feature to the inputs (e.g., 1, 3, 5 copies of energy)
  - **Draw a reliability diagram**, which groups the validation tracks by their confidence (e.g., 0 to 0.1, 0.1 to 0.2, …, 0.9 to 1), and plot the accuracy of each group against its average confidence.
    - *The confidence on a track is the predicted probability of the genre the classifier predicts (the largest posterior probability).*

Decision Trees and Their Ensembles (9%)

- **Look inside the prediction**
  - Train a tree with a maximum depth of 3 and plot it. Describe the feature and threshold of each split.
  - Explain why the tree splits on these first, using **how a split is chosen by measuring the decrease in *impurity***.
  - List the genres that such a small tree never predicts, and explain why.
- **Growing and Pruning**
  - Vary **at least two hyperparameters** that control the size of the tree (e.g., maximum depth, minimum samples per leaf), each over a range wide enough to **go from clear underfitting to clear overfitting**.

Decision Trees and Their Ensembles (9%)

- **Many Trees**
  - Compare **a single fully grown tree, bagging, and a random forest.**
  - Measure how much each model's predictions change when it is trained on a different random sample.
  - Show how the scores of bagging and of the random forest change with the **number of trees from 1 to at least a few *hundred*.**
  - Explain your observations about **why averaging many models helps**, and **compare the random forest with bagging** both in score and in how much their predictions change with **mathematical evidence**.

Support Vector Machines (9%)

- The two **hyperparameters of an RBF kernel**
  - Evaluate an **RBF-kernel** SVM on standardized features over a grid of ***C* and γ.**
  - Each spanning at least four orders of magnitude.
  - **Identify the regions where the model underfits and where it overfits.**
- **Support Vectors**
  - Fit the SVM on the whole training set and report the fraction of training tracks for each genre that become support vectors. Relate the fraction of each genre to **how well that genre is recognized (via cross-validated F_1)** and to **how many training tracks** it has.

Section 4: Beyond Accuracy (10%)

- **Different numbers, different conclusions**
  - Using the best setting you found for each classifier family, report the cross-validated accuracy and macro-F_1 of every family.
  - For your best model (state by which metric), showed how its **multi-class confusion matrix** on the test set.
  - Compute a **one-versus-rest ROC-AUC** for every genre from the model's predicted probabilities on the test set.

Section 4: Beyond Accuracy (10%)

- *The team also wants a detector that answers "hip-hop" or "not hip-hop" for each track:*
  - Build detectors from **at least two families** that output a continuous score (e.g., a probability or a decision value).
  - Train them on **two-class labels** (hip-hop or not), and choose their hyperparameters without the test set.
  - For each detector at its default threshold (a probability of 0.5, or a decision value of 0), report **the confusion matrix, accuracy, TPR, TNR, PPV, NPV, and F_1 .**
  - Plot the **ROC and precision-recall curves**, and report the area under each.
  - A curator will listen to every track the detector flags, and wants to find **at least 80%** of the hip-hop tracks. For one of your detectors, **choose a threshold** that meets this need. **On the test set, report whether the recall reaches 80% and what it costs** (e.g., tracks to listen to and hip-hop tracks missed per 1,000 tracks).

Grading Policy

- **Report 50% + Demo 50%**
- To prevent grader bias, each section is assigned to a dedicated TA to grade.

Tier	Description	Percentage
4	Provides evidence, explanation, and decisions, along with a discussion of limitations or alternative interpretations.	100%
3	Provides evidence, explanation, and decisions, with coherent logic linking all three.	80%
2	Presents results or figures, but the explanation is weak or disconnected from the evidence.	50%
1	Only presents results without explanation, or the methodology is clearly flawed.	20%
0	No answer provided / Blank.	0%

Appendix

- Demo will take place on **Thursday, November 12**, together with the HW1 demo.
- Demo details will be announced later.
- For homework-related issues, please **post on the eeclass discussion forum**, and the TAs will reply ASAP.
- For personal matters or private inquiries, you can reach the TA at george930502@gapp.nthu.edu.tw.

Q & A